Week 13: Target Trial Emulation

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STAT 579 – Applied Causal Inference: Target Trial Emulation

Overview

This lecture introduces the **target trial framework**, a unifying conceptual structure for designing and critiquing observational research by explicitly emulating a hypothetical randomized trial. Unlike earlier discussions of simplified target trials comparing time-fixed treatments, this session addresses realistic trials comparing **sustained treatment strategies** over time. We explore how to define, identify, and estimate causal effects in both randomized experiments and observational studies, emphasizing that both require similar methodological rigor when dealing with time-varying treatments, non-adherence, and loss to follow-up.

The target trial framework bridges counterfactual reasoning and causal diagrams, grounding abstract theory in the familiar concept of the experiment. By specifying exactly which randomized trial we would conduct if we could, we clarify our causal question, identify the interventions we wish to compare, and determine the data and methods needed for valid inference. This approach reveals that per-protocol analyses of randomized trials and observational studies face identical challenges—time-varying confounding and selection bias—requiring g-methods for valid estimation (Hernán & Robins, 2020).

Learning Objectives

By the end of this session, students will be able to:

1. **Articulate** the distinction between intention-to-treat (ITT) and per-protocol effects in both randomized trials and observational studies, including precise counterfactual definitions of each
2. **Specify** the key components of a target trial protocol (eligibility, time zero, treatment strategies, outcomes, causal contrast, analysis plan) for a given research question
3. **Identify** common time-zero errors in observational studies and **apply** solutions including sequential trial emulation and cloning-censoring-weighting procedures
4. **Recognize** when conventional ITT and "per-protocol" analyses fail to estimate their intended causal effects and **justify** the need for g-methods
5. **Design** an observational analysis that explicitly emulates a target trial, including handling multiple eligibility times and non-unique strategy assignment

R Packages

```
pacman::p_load(
  tidyverse, # data manipulation and visualization
  dagitty, # DAG analysis
  ggdag, # DAG visualization
  gt, # tables
  gtsummary, # summary tables
  survival, # survival analysis
  broom, # tidy model outputs
  knitr, # document generation
  kableExtra # enhanced tables
)
```

1 Comprehensive Topic Explanation

1.1 From Simple to Realistic Target Trials

Throughout Parts I and II of *What If*, we considered simplified target trials comparing **time-fixed treatments**—interventions assigned at baseline with perfect adherence. While pedagogically useful, this simplification obscures critical features of real-world research (Hernán & Robins, 2020, Chapter 22). Realistic trials compare **sustained treatment strategies** that unfold over time, permit protocol deviations (non-adherence, treatment switching), and experience loss to follow-up.

Consider a trial evaluating antiretroviral therapy in HIV-infected individuals. A realistic protocol would not simply assign "treatment" versus "no treatment" at baseline and assume everyone adheres perfectly for 5 years. Instead, it would specify:

- **Strategy** g_1: "Initiate therapy immediately and continue unless contraindications or toxicity arise"
- **Strategy** g_0: "Defer therapy indefinitely"

These are **dynamic strategies** that incorporate decision rules responding to evolving patient status. The first strategy mandates treatment discontinuation if medically necessary—this is adherence to the protocol, not deviation from it.

1.2 Intention-to-Treat vs. Per-Protocol Effects

Defining ITT Effects

The **intention-to-treat (ITT) effect** compares outcomes under assignment to different strategies at baseline, regardless of subsequent adherence. Let Z denote randomized assignment ($Z = 1$ for strategy g_1, $Z = 0$ for strategy g_0), A_k denote received treatment at time k, and D_k denote death by time k. The ITT effect at time k is:

$$\text{ITT Effect}_k = \Pr[D_k^{z=1,\bar{c}_k=\bar{0}} = 1] - \Pr[D_k^{z=0,\bar{c}_k=\bar{0}} = 1]$$

where $\bar{c}_k = \bar{0}$ indicates no censoring through time k. This contrast reflects:

1. The effect of **assignment** to treatment strategies
2. The effect of **initiating** treatment (if assignment and initiation coincide)
3. All downstream consequences of assignment, including adherence patterns, behavioral changes, and awareness effects

> **! Key Distinction**
>
> The ITT effect is **not** "the effect of treating with A" but rather "the effect of assigning participants to being treated with A" or "the effect of having the intention of treating with A."

The causal effect of assignment Z depends on three pathways (see Figure 22.1 in the text):

1. $Z \to A \to Y$: Assignment affects received treatment, which affects outcome
2. $Z \to Y$: Assignment directly affects outcome (e.g., through awareness, behavioral changes)
3. The strength of $Z \to A$ (adherence level in the study)

Defining Per-Protocol Effects

The **per-protocol effect** compares outcomes under **full adherence** to the assigned strategies:

$$\text{Per-protocol Effect}_k = \Pr[D_k^{g_1,\bar{c}_k=\bar{0}} = 1] - \Pr[D_k^{g_0,\bar{c}_k=\bar{0}} = 1]$$

This is the effect that would be observed if everyone followed their assigned strategy as specified in the protocol. For sustained strategies, this generally involves comparing **dynamic** rather than static strategies.

⚠ Critical Insight

Estimating per-protocol effects requires viewing the randomized experiment as an **observational study**. Non-adherence creates confounding via the backdoor path $A \leftarrow U \rightarrow Y$, where U represents unmeasured factors affecting both adherence and outcome.

1.3 Why ITT Effects Are Insufficient

Several justifications for privileging ITT effects require scrutiny:

Null Preservation: Under the sharp causal null and exclusion restriction, if A has no effect on Y, then Z has no effect on Y. However, this fails when the exclusion restriction doesn't hold (non-blinded trials). The arrow $Z \rightarrow Y$ can exist even when $A \rightarrow Y$ is absent.

Conservativeness: The claim that ITT effects provide lower bounds for per-protocol effects (i.e., are "conservative") is not universally true. With high non-adherence or non-monotonic effects, the per-protocol effect may be closer to the null than the ITT effect. In head-to-head trials of two active treatments, differential adherence can make the ITT risk ratio further from the null than the per-protocol risk ratio.

Effectiveness vs. Efficacy: This dichotomy is misleading. The ITT effect measures the effect of assignment under the specific adherence conditions in *that particular trial*—conditions that may differ in other settings or change after evidence emerges. The per-protocol effect is often more policy-relevant because decision-makers want to know the effect of actually following a treatment strategy.

1.4 Target Trials with Sustained Strategies

Consider a trial randomizing HIV-infected individuals (18+, no AIDS, no prior antiretroviral therapy) to:

- **Strategy** g_1: Receive $A_k = 1$ continuously unless contraindications/toxicity arise
- **Strategy** g_0: Receive $A_k = 0$ continuously

Follow-up starts at randomization ($k = 0$) and continues for 60 months or until death/loss to follow-up.

The ITT effect contrasts static strategies:

- $(z = 1, \bar{c}_K = \bar{0})$: Assigned to g_1 and remain in study
- $(z = 0, \bar{c}_K = \bar{0})$: Assigned to g_0 and remain in study

The per-protocol effect contrasts dynamic strategies:

- Receive strategy g_1 continuously (including medically-indicated discontinuations)
- Receive strategy g_0 continuously

> Protocol Clarity
>
> Sensible protocols mandate treatment continuation only while medically appropriate. Discontinuing therapy due to toxicity is **adherence** to the protocol, not deviation. Protocols should explicitly define what constitutes adherence versus deviation.

1.5 Emulating Target Trials with Observational Data

When randomized trials are infeasible, we emulate them using observational data. Specifying the target trial protocol forces us to clarify:

1. **Eligibility criteria**: Who would be enrolled?
2. **Time zero**: When does follow-up start?
3. **Treatment strategies**: What interventions are compared?
4. **Outcomes**: What is measured and when?
5. **Causal contrast**: ITT analog, per-protocol, or other?
6. **Analysis plan**: How will effects be estimated?

Observational Analog of ITT: Rarely possible because actual assignment is unknown. The closest analog compares **initiators** of different strategies at baseline:

$$\Pr[D_k^{a_0=1,\bar{c}_k=\bar{0}} = 1] - \Pr[D_k^{a_0=0,\bar{c}_k=\bar{0}} = 1]$$

This resembles a modified ITT analysis, defining groups by treatment initiation rather than assignment.

Observational Analog of Per-Protocol: Defined identically to the randomized trial per-protocol effect. We can only emulate target trials whose interventions are actually followed by some individuals in the data (or use structural models to extrapolate).

1.6 The Time-Zero Problem

Time zero (baseline) is when:

- Eligibility criteria are met
- Treatment strategies are assigned (in trials) or can be assigned (in observational studies)
- Follow-up begins
- Outcomes start being counted

Errors in defining time zero are endemic in observational research and lead to two major problems:

Problem 1: Non-Unique Time Zero

Eligibility criteria may be met at **multiple times** during an individual's life. Example: A study of hormone therapy in postmenopausal women with no chronic disease. If a woman meets criteria continuously from age 51-65, when is her time zero?

Solutions:

a) Choose first eligible time
b) Choose randomly among eligible times

c) **Sequential trial emulation**: Emulate multiple trials, one starting at each eligible time

Strategy (c) is most efficient but requires:

- Fixed time units (days, weeks, months) if data collection is subject-specific (e.g., medical records)
- Pre-specified times (e.g., study visits) if data collection follows a fixed schedule
- Bootstrapping to account for individuals appearing in multiple trials

Problem 2: Non-Unique Strategy Assignment

At baseline, an individual's observed data may be compatible with multiple strategies. Example: Comparing three strategies based on CD4 thresholds (initiate immediately, at <350, or at <200). An individual who doesn't initiate immediately has data consistent with both delay strategies.

Solution: Cloning + Censoring + Weighting

1. **Clone** the individual, creating copies assigned to each compatible strategy
2. **Censor** each clone when their data become incompatible with their assigned strategy
3. **Weight** to adjust for informative censoring using IP weights
4. **Bootstrap** to account for correlated data

Example: If someone doesn't initiate at CD4=350, the clone assigned to "initiate at <350" is censored. If they died before any clone was censored, the death is assigned to all strategies.

1.7 Unified Framework for Causal Inference

The target trial framework provides a **unified approach** to causal inference:

1. **Common language**: Same concepts apply to trials and observational studies

2. **Same methods**: G-methods handle time-varying confounding in both settings
3. **Same challenges**: Selection bias from loss to follow-up affects ITT estimates in trials; confounding affects per-protocol estimates in both

The key insight: **A randomized trial is a follow-up study with baseline randomization; an observational study is a follow-up study without baseline randomization**. For per-protocol effects of sustained strategies, both face identical methodological challenges.

2 Intuitive Clarifications of Challenging Ideas

2.1 Why Isn't ITT Always Conservative?

The Intuition: Many believe imperfect adherence always dilutes treatment effects, making ITT estimates "conservative" (closer to the null than per-protocol effects).

The Reality: Consider a head-to-head trial of Drug A vs. Drug B for pain relief (lower is better). Unknown to investigators, both drugs are equally effective (per-protocol risk ratio = 1). But Drug B has mild side effects causing more people to stop it. The ITT risk ratio > 1 wrongly suggests Drug B is inferior—the ITT effect is further from the null than the true per-protocol effect.

The Lesson: "Conservative" only means "systematically biased in one direction"—which isn't necessarily better than unbiased.

2.2 The Exclusion Restriction Intuition

The **exclusion restriction** states: Assignment Z affects outcome Y **only through** received treatment A, with no direct arrow $Z \to Y$.

Visual Metaphor: Think of Z as a switch that only controls the treatment machine A. If flipping the switch changes your outcome through any other pathway (awareness, psychological effects, behavioral changes), the exclusion restriction fails.

When It Holds: Double-blind placebo-controlled trials attempt to eliminate $Z \to Y$ by making assignment unobservable to participants and clinicians.

When It Fails: Most pragmatic trials, all observational studies, trials where treatment has obvious side effects.

2.3 Why Per-Protocol Analysis Is Observational

The Paradox: You carefully randomized 1000 people. Why does analyzing those who adhered require treating the data as observational?

The Explanation: Randomization ensures $Z \perp!!! \perp Y^z$ (assigned groups are exchangeable). But adherence is **post-randomization** and typically related to prognosis:

- High-risk patients in the control arm may seek treatment outside the study
- Patients experiencing side effects may stop treatment
- Healthier patients may adhere better

These create confounding: $A \leftarrow U \to Y$. The randomization cannot eliminate U because U affects decisions made *after* randomization.

The Analogy: Imagine randomizing students to study groups, but some students skip group meetings. Comparing outcomes by actual attendance (not assignment) requires accounting for factors that predict both attendance and performance.

2.4 Cloning Intuition

Why Clone? When someone's baseline data are compatible with multiple strategies, randomly assigning them to one strategy wastes information.

The Solution: Create clones assigned to each compatible strategy. Track each clone until its observed behavior contradicts its assigned strategy, then censor.

Key Insight: If death occurs before any clone is censored, the death is assigned to **all** strategies. This prevents systematically assigning early events to only one strategy (which would bias results).

The Cost: Clones are correlated, so we must bootstrap to get valid standard errors.

2.5 Why G-Methods Are Necessary

For Per-Protocol Effects: Non-adherence creates time-varying confounding. At each time k:

- Past treatment affects current confounders: $A_{k-1} \to L_k$
- Current confounders affect current treatment: $L_k \to A_k$
- Current confounders affect outcome: $L_k \to Y$

Standard regression cannot properly adjust because L_k is simultaneously a confounder (must condition on) and a mediator (must not condition on). G-methods (IP weighting, g-formula, g-estimation) resolve this.

For ITT Effects with LTFU: Loss to follow-up creates selection bias. The probability of remaining in the study depends on time-varying prognostic factors. G-methods adjust for this selection.

3 Simple, Hand-Calculation Numerical Example(s)

3.1 Example: ITT Vs Per-Protocol with Non-Adherence

Consider a trial of vaccine ($Z = 1$) vs. no vaccine ($Z = 0$) to prevent infection. Each individual has a risk type U:

- $U = 1$ (high-risk): 50% of population
- $U = 0$ (low-risk): 50% of population

True vaccine effect (per-protocol):

- For high-risk: $P(Y^{a=1} = 1) = 0.2$, $P(Y^{a=0} = 1) = 0.6$ (Risk Difference = -0.4)

- For low-risk: $P(Y^{a=1} = 1) = 0.1$, $P(Y^{a=0} = 1) = 0.3$ (Risk Difference = -0.2)

Trial design: Randomize 200 people (100 per arm). Perfect adherence in $Z = 1$ arm. In $Z = 0$ arm, all high-risk people seek vaccination outside the study ($A = 1$), while low-risk people remain unvaccinated ($A = 0$).

Worked Example: Computing ITT and Per-Protocol Effects

Step 1: Calculate observed outcomes

$Z = 1$ *arm (all receive vaccine):* - 50 high-risk: $50 \times 0.2 = 10$ infected - 50 low-risk: $50 \times 0.1 = 5$ infected - Total: 15/100 infected

$Z = 0$ *arm:* - 50 high-risk get vaccine: $50 \times 0.2 = 10$ infected - 50 low-risk remain unvaccinated: $50 \times 0.3 = 15$ infected - Total: 25/100 infected

Step 2: ITT effect (intention-to-treat)

$$\text{ITT RD} = P(Y = 1|Z = 1) - P(Y = 1|Z = 0) = 0.15 - 0.25 = -0.10$$

Step 3: Per-protocol effect (marginal average)

$$\text{PP RD} = 0.5 \times (-0.4) + 0.5 \times (-0.2) = -0.30$$

Step 4: Naive per-protocol analysis (compares $A = 1$ vs $A = 0$)

- $A = 1$: 150 people (all high-risk from $Z = 0$, all from $Z = 1$), $20/150 = 0.133$ infected
- $A = 0$: 50 people (low-risk from $Z = 0$), $15/50 = 0.30$ infected
- Naive PP RD $= 0.133 - 0.30 = -0.167$ (biased!)

Interpretation:

- **ITT effect** (-0.10) is attenuated because high-risk controls got vaccinated
- **Per-protocol effect** (-0.30) is the true average causal effect
- **Naive PP estimate** (-0.167) is biased because $A = 1$ group has more high-risk people

Key Lesson: The ITT effect is easy to compute but may not answer the right question. The per-protocol effect requires adjustment for post-randomization confounding.

3.2 Example: Time-Zero Error

Consider studying whether statin initiation reduces cardiovascular events in people with diabetes.

Correct approach: Time zero = diabetes diagnosis. Compare:

- Group A: Initiate statin within 30 days of diagnosis
- Group B: Never initiate statin

Incorrect approach 1 (immortal time bias): Define time zero as statin initiation for Group A, but also at diabetes diagnosis for Group B.

Person	Diabetes Dx	Statin Start	Death	Assignment	Follow-up Start
1	Jan 2020	Mar 2020	-	Statin	Mar 2020
2	Jan 2020	-	Feb 2020	No Statin	Jan 2020

Person 1 cannot die between Jan-Mar (immortal time). Person 2 can die immediately. This guarantees finding a "protective" effect of statins even if they're harmful!

Incorrect approach 2 (prevalent user bias): Compare current statin users to non-users, ignoring duration of use and indication.

Common Pitfall

Never start follow-up at different calendar times for different strategy groups unless you're explicitly emulating sequential trials with different start times.

4 R Code Examples & Interpretation

4.1 Visualizing ITT Vs Per-Protocol Effects

```
# Create DAG for per-protocol confounding
dag <- dagify(
  Y ~ A + U + Z,
  A ~ Z + U,
  exposure = "A",
  outcome = "Y",
  coords = list(
    x = c(Z = 1, U = 2, A = 3, Y = 4),
    y = c(Z = 2, U = 1, A = 2, Y = 2)
  )
)

# Plot with ggdag
ggdag(dag, text_size = 3.5) +
  theme_dag() +
  labs(
    title = "Per-Protocol Confounding",
    subtitle = "Backdoor path A ← U → Y creates
    confounding"
  )
```

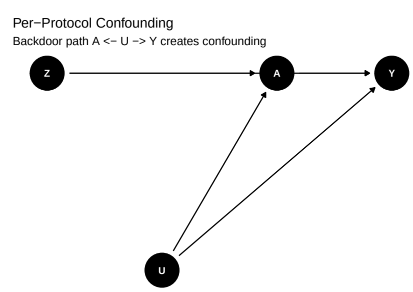

Figure 1: Causal diagram showing confounding for per-protocol effect

Interpretation: Even though Z is randomized ($Z \perp!!! \perp Y^z$), the received treatment A is not randomized relative to Y due to the common cause U. This necessitates adjustment for U (or proxies of U) when estimating per-protocol effects.

4.2 Simulating a Trial with Non-Adherence

```
set.seed(579)
n <- 1000

# Simulate trial data
trial_data <- tibble(
  id = 1:n,
  # Unmeasured risk factor
  U = rbinom(n, 1, 0.5),
  # Randomized assignment
  Z = rbinom(n, 1, 0.5),
  # Received treatment (adherence depends on U and Z)
  A = case_when(
    Z == 1 ~ 1, # Perfect adherence in treatment arm
    Z == 0 & U == 1 ~ rbinom(n, 1, 0.7), # High-risk
seek treatment
    Z == 0 & U == 0 ~ 0, # Low-risk remain untreated
    TRUE ~ 0 # Default case
  ),
  # Outcome (depends on A and U)
  Y = rbinom(
    n,
    1,
    prob = case_when(
      A == 1 & U == 1 ~ 0.20,
      A == 1 & U == 0 ~ 0.10,
      A == 0 & U == 1 ~ 0.60,
      A == 0 & U == 0 ~ 0.30
    )
  )
)

# Calculate effects
trial_summary <- trial_data %>%
  summarise(
    # ITT effect
```

```r
    ITT_Risk_Z1 = mean(Y[Z == 1]),
    ITT_Risk_Z0 = mean(Y[Z == 0]),
    ITT_RD = ITT_Risk_Z1 - ITT_Risk_Z0,

    # Naive per-protocol (unadjusted)
    PP_Risk_A1 = mean(Y[A == 1]),
    PP_Risk_A0 = mean(Y[A == 0]),
    Naive_PP_RD = PP_Risk_A1 - PP_Risk_A0,

    # True per-protocol (adjusted by U, but U is
    unmeasured)
    # We can calculate it because we simulated U
    True_PP_RD = 0.5 * (-0.4) + 0.5 * (-0.2)
  )

trial_summary %>%
  pivot_longer(everything(), names_to = "Measure",
  values_to = "Value") %>%
  gt() %>%
  fmt_number(columns = Value, decimals = 3) %>%
  tab_header(title = "Comparison of ITT and Per-Protocol
  Estimates")
```

Interpretation:

- The **ITT effect** is attenuated (closer to null) because many controls got treated
- The **naive per-protocol estimate** is biased because the $A = 1$ group includes more high-risk individuals
- The **true per-protocol effect** (-0.30) requires adjustment for U, which is unmeasured in real data

4.3 Sequential Trial Emulation

```r
# Simulate cohort with multiple eligible times
```

Table 2: Simulated trial results by randomization and adherence

[

Comparison of ITT and Per-Protocol Estimates] Comparison of ITT and Per-Protocol Estimates

Measure	Value
ITT_Risk_Z1	0.177
ITT_Risk_Z0	0.293
ITT_RD	-0.116
PP_Risk_A1	0.181
PP_Risk_A0	0.349
Naive_PP_RD	-0.168
True_PP_RD	-0.300

```r
set.seed(579)
cohort <- expand_grid(
  id = 1:200,
  time = 0:5
) %>%
  group_by(id) %>%
  mutate(
    # Eligibility (e.g., postmenopausal, no chronic
    disease)
    eligible = time ≥ rbinom(1, 5, 0.5),
    # Treatment initiation
    initiated = if_else(eligible, rbinom(1, 1, 0.3), 0),
    # Once initiated, stays initiated
    A = cummax(initiated),
    # Outcome risk increases with time and is lower with
    treatment
    Y = rbinom(1, 1, prob = 0.05 * time - 0.02 * A)
  ) %>%
  ungroup()
```

```r
# Emulate trials starting at each time point
trial_results <- cohort %>%
  filter(eligible) %>%
  group_by(time) %>%
  nest() %>%
  mutate(
    trial_data = map(
      data,
      ~ {
        # For each time, create trial
        .x %>%
          filter(time == min(time)) %>% # Baseline for this trial
          left_join(
            cohort %>%
              filter(time > min(.x$time)) %>%
              select(id, time, A, Y),
            by = "id"
          )
      }
    ),
    # Estimate effect for each trial
    effect = map_dbl(
      trial_data,
      ~ {
        mean(.x$Y[.x$A == 1], na.rm = TRUE) -
          mean(.x$Y[.x$A == 0], na.rm = TRUE)
      }
    )
  )

# Pool across trials (simple average; proper analysis would use bootstrapping)
trial_results %>%
  summarise(
    n_trials = n(),
    pooled_effect = mean(effect, na.rm = TRUE),
```

[

Pooled Results from Sequential Trial Emulation] Pooled Results from Sequential Trial Emulation

time	n_trials	pooled_effect	se_effect
0.000	1.000	NaN	NA
1.000	1.000	NaN	NA
2.000	1.000	NaN	NA
3.000	1.000	NaN	NA
4.000	1.000	NaN	NA
5.000	1.000	NaN	NA

```
    se_effect = sd(effect, na.rm = TRUE) / sqrt(n())
  ) %>%
  gt() %>%
  fmt_number(decimals = 3) %>%
  tab_header(title = "Pooled Results from Sequential
  Trial Emulation")
```

Interpretation: By emulating multiple trials (one starting at each time individuals become eligible), we use more available data and avoid arbitrary choice of a single time zero. Proper implementation requires bootstrapping to account for individuals appearing in multiple trials.

4.4 Cloning for Non-Unique Strategy Assignment

```
# Simulate individuals whose baseline data are
compatible with multiple strategies
set.seed(579)
base_data <- tibble(
  id = 1:100,
  cd4_baseline = rnorm(100, 400, 100)
```

```
) %>%
  filter(cd4_baseline > 300 & cd4_baseline < 500) #
  Eligible for all strategies

# Strategies:
# g1: Initiate immediately
# g2: Initiate when CD4 < 350
# g3: Initiate when CD4 < 200

# Clone each individual into 3 copies
cloned_data <- base_data %>%
  crossing(strategy = c("Immediate", "At_350",
  "At_200")) %>%
  mutate(clone_id = paste0(id, "_", strategy))

# Simulate follow-up
set.seed(579)
followup <- cloned_data %>%
  crossing(month = 0:12) %>%
  group_by(clone_id) %>%
  mutate(
    # CD4 declines over time (simplified)
    cd4 = cd4_baseline - month * 15 + rnorm(n(), 0, 20),

    # Observed treatment (person's actual behavior, same
    across clones)
    observed_treatment = case_when(
      month == 0 ~ 0, # Nobody treated at baseline
      month <= 3 ~ 0, # Delay 3 months
      TRUE ~ 1 # Then initiate
    ),

    # Treatment assigned by strategy
    assigned_treatment = case_when(
      strategy == "Immediate" ~ if_else(month >= 0, 1,
0),
      strategy == "At_350" ~ if_else(cd4 < 350, 1, 0),
```

```
      strategy == "At_200" ~ if_else(cd4 < 200, 1, 0)
    ),

    # Censor when observed != assigned
    censored = observed_treatment != assigned_treatment,
    censored = cumsum(censored) > 0,

    # Outcome
    death = if_else(
      !censored,
      rbinom(n(), 1, 0.01 + 0.02 * (1 -
      observed_treatment)),
      NA_integer_
    )
  ) %>%
  ungroup()

# Estimate effect (simplified; would need IP weighting
for censoring)
followup %>%
  filter(!censored, month == 12) %>%
  group_by(strategy) %>%
  summarise(
    n = n(),
    deaths = sum(death, na.rm = TRUE),
    risk = mean(death, na.rm = TRUE)
  ) %>%
  gt() %>%
  fmt_number(columns = risk, decimals = 3) %>%
  tab_header(title = "Risk by Strategy (Cloning
  Approach)")
```

Interpretation: Each individual is cloned into copies assigned to each strategy they could follow at baseline. Clones are censored when observed behavior contradicts assigned strategy. This avoids arbitrarily assigning individuals to a single strategy and throwing away information.

[

Risk by Strategy (Cloning Approach)] Risk by Strategy (Cloning Approach)

strategy	n	deaths	risk
At_350	1	0	0.000

5 Hands-On R Lab

Lab Objectives

1. Specify a target trial protocol for a given research question
2. Identify and classify protocol deviations as adherence vs. non-adherence
3. Implement sequential trial emulation for multiple eligibility times
4. Apply cloning for non-unique strategy assignment
5. Compare ITT and per-protocol analyses

Pre-Lab Reading

Before lab, review:

- Hernán & Robins (2020), Chapter 22, Sections 22.1-22.4
- The concepts of ITT effect, per-protocol effect, and time zero

Lab Scenario

You have electronic health record data on 5,000 patients with newly diagnosed type 2 diabetes between 2015-2020. You want to estimate the 3-year risk of cardiovascular events under two treatment strategies:

- **Intensive strategy**: Initiate ≥2 medications immediately
- **Standard strategy**: Initiate 1 medication, add others as needed

Task 1: Specify the Target Trial Protocol

Write out the key components of your target trial:

```
# Target Trial Protocol Specification
# -----------------------------------

# 1. Eligibility criteria:
# Adults (18+) with new type 2 diabetes diagnosis
# No prior diabetes medications
# No cardiovascular disease at diagnosis

# 2. Time zero:
# Date of diabetes diagnosis

# 3. Treatment strategies:
# g1 (intensive): Initiate ≥2 medications within 30 days
of diagnosis
# g0 (standard): Initiate 1 medication within 30 days,
may add others later

# 4. Outcome:
# Cardiovascular event (MI, stroke, CV death) within 36
months

# 5. Causal contrast:
# Per-protocol effect: Risk difference under full
adherence to g1 vs g0

# 6. Analysis plan:
# Sequential trial emulation with monthly time units
# IP weighting for time-varying confounding and
censoring
```

Task 2: Simulate Data and Identify Time Zero

```r
set.seed(20241111)

# Simulate patient cohort
n_patients <- 500 # Reduced for lab

diabetes_cohort <- tibble(
  patient_id = 1:n_patients,

  # Demographics
  age = rnorm(n_patients, 55, 10),
  female = rbinom(n_patients, 1, 0.45),

  # Diagnosis date (time zero!)
  dx_date = as.Date("2015-01-01") + runif(n_patients, 0,
  365 * 5),

  # Baseline risk factors
  hba1c_baseline = rnorm(n_patients, 8.5, 1.5),
  sbp_baseline = rnorm(n_patients, 140, 20),

  # Treatment initiation
  days_to_treat = rexp(n_patients, 1 / 60) * (1 + 0.5 *
  (hba1c_baseline > 9)),

  # Number of initial medications (strategy assignment)
  n_meds_initial = case_when(
    days_to_treat <= 30 & hba1c_baseline > 9 ~
rpois(n_patients, 2),
    days_to_treat <= 30 ~ 1,
    TRUE ~ 0
  )
) %>%
  mutate(
    strategy = case_when(
      n_meds_initial >= 2 ~ "Intensive",
```

[

Time Zero = Date of Diabetes Diagnosis] Time Zero = Date of Diabetes Diagnosis

patient_id	dx_date	days_to_treat	n_meds_initial	strategy
1	17306	67.651693	0	None
2	18173	4.645261	3	Intensive
3	17731	83.345429	0	None
4	17967	57.407807	0	None
5	17683	46.149661	0	None
6	16444	166.822424	0	None
7	17676	65.184190	0	None
8	18216	19.715412	1	Standard
9	16818	143.509531	0	None
10	16620	140.933966	0	None

```
      n_meds_initial == 1 ~ "Standard",
      TRUE ~ "None"
    )
  )

# Inspect time zero
diabetes_cohort %>%
  select(patient_id, dx_date, days_to_treat,
  n_meds_initial, strategy) %>%
  head(10) %>%
  gt() %>%
  tab_header(title = "Time Zero = Date of Diabetes
  Diagnosis")
```

Check Your Understanding

Q1: Why is `dx_date` the correct time zero? **A1**: It's when eligibility criteria are met (new diabetes diagnosis, no prior meds), and when treatment strategies would be assigned in the target trial.

Q2: What would be wrong with using `days_to_treat` as time zero?
A2: This varies across patients and occurs *after* diagnosis. Starting follow-up at treatment initiation would: - Exclude patients who died before treatment - Miss early treatment effects - Create selection bias if sicker patients delay treatment

Task 3: Implement Basic Intention-to-Treat Analog

```r
# ITT analog: Compare initiators at baseline
itt_data <- diabetes_cohort %>%
  filter(days_to_treat <= 30) %>% # Initiated within grace period
  mutate(
    # Simulate 3-year outcomes
    cvd_risk_3y = case_when(
      strategy == "Intensive" ~ 0.08 +
        0.01 * (age - 55) / 10 +
        0.03 * (female == 0),
      strategy == "Standard" ~ 0.12 +
        0.01 * (age - 55) / 10 +
        0.03 * (female == 0),
      TRUE ~ 0.15
    ),
    cvd_event = rbinom(n(), 1, cvd_risk_3y)
  )

# Estimate ITT analog effect
itt_results <- itt_data %>%
  group_by(strategy) %>%
  summarise(
    n = n(),
    events = sum(cvd_event),
    risk = mean(cvd_event)
  ) %>%
  arrange(desc(strategy))
```

[

ITT Analog: Comparing Treatment Initiators] ITT Analog: Comparing Treatment Initiators

strategy	n	events	risk
Standard	149	23	0.154
None	5	0	0.000
Intensive	42	7	0.167

```
itt_results %>%
  gt() %>%
  fmt_number(columns = risk, decimals = 3) %>%
  tab_header(title = "ITT Analog: Comparing Treatment
  Initiators")
```

```
# Risk difference
risk_diff <- itt_results$risk[1] - itt_results$risk[2]
cat("ITT Analog Risk Difference:", round(risk_diff, 3),
"\n")
```

```
ITT Analog Risk Difference: 0.154
```

Student Exercise: Modify the code to:

a) Add adjustment for age and sex using logistic regression
b) Calculate a confidence interval for the risk difference
c) Interpret why this is only an *analog* of the true ITT effect

```
# Solution: Adjusted ITT analog

# Fit logistic regression
model_adj <- glm(
```

```r
  cvd_event ~ strategy + age + female,
  data = itt_data,
  family = binomial()
)

# Extract adjusted estimates (marginal standardization)
adjusted_risks <- itt_data %>%
  mutate(
    pred_intensive = predict(
      model_adj,
      newdata = mutate(., strategy = "Intensive"),
      type = "response"
    ),
    pred_standard = predict(
      model_adj,
      newdata = mutate(., strategy = "Standard"),
      type = "response"
    )
  ) %>%
  summarise(
    risk_intensive = mean(pred_intensive),
    risk_standard = mean(pred_standard),
    adj_RD = risk_intensive - risk_standard
  )

adjusted_risks %>%
  gt() %>%
  fmt_number(decimals = 3) %>%
  tab_header(title = "Adjusted ITT Analog Effect")
```

```r
# Bootstrap CI
library(boot)

boot_rd <- function(data, indices) {
  d <- data[indices, ]
  m <- glm(cvd_event ~ strategy + age + female, data =
d, family = binomial())
```

[

Adjusted ITT Analog Effect]

Adjusted ITT Analog Effect

risk_intensive	risk_standard	adj_RD
0.168	0.154	0.014

```
  d_aug <- d %>%
    mutate(
      pred_int = predict(
        m,
        newdata = mutate(d, strategy = "Intensive"),
        type = "response"
      ),
      pred_std = predict(
        m,
        newdata = mutate(d, strategy = "Standard"),
        type = "response"
      )
    )
  mean(d_aug$pred_int) - mean(d_aug$pred_std)
}

boot_results <- boot(itt_data, boot_rd, R = 500)
boot_ci <- boot.ci(boot_results, type = "perc")

cat(
  "\n95% Bootstrap CI:",
  round(boot_ci$percent[4], 3),
  "to",
  round(boot_ci$percent[5], 3),
  "\n"
)
```

```
95% Bootstrap CI: -0.11 to 0.146
```

Task 4: Sequential Trial Emulation

```r
# Create person-month dataset
person_months <- diabetes_cohort %>%
  crossing(month = 0:36) %>%
  group_by(patient_id) %>%
  mutate(
    # Follow-up time
    time_since_dx = month,

    # Still eligible (simplified: no CVD yet, no death)
    eligible = month == 0 | lag(cvd_event, default = 0)
    == 0,

    # Treatment assignment at this time point
    # (In reality, would determine based on observed
    treatment patterns)

    # Outcome in this month
    cvd_event = if_else(
      eligible,
      rbinom(n(), 1, 0.002 * (1 + 0.5 * (strategy ==
      "Standard"))),
      NA_real_
    )
  ) %>%
  ungroup()

# YOUR TURN: Complete the sequential trial emulation
# For each month, create a trial including all eligible
patients
# Follow them forward and estimate effect

# Hint: Use nest() and map() to iterate over start times
```

Implementation Hint

For each potential time zero (month 0, 1, 2, ..., 36):

1. Filter to patients eligible at that month
2. Assign strategies based on treatment within next 30 days
3. Follow forward to measure 36-month outcome
4. Estimate effect for this trial
5. Pool across trials using bootstrapping

Task 5: Reflection and Discussion

After completing the lab, discuss:

1. **When would you prefer ITT analog vs. per-protocol effect?** What factors determine which is more policy-relevant?
2. **How would you handle grace periods?** If we allow 30 days to initiate "immediate" treatment, how does this affect cloning and censoring?
3. **What if eligibility changes over time?** Suppose some patients develop CVD (making them ineligible). How does this affect sequential trial emulation?
4. **Unmeasured confounding**: What patient characteristics might affect both treatment choice and outcomes but aren't in EHR data?

Post-Lab Quiz

Quick Knowledge Check

Q1: True or False: The ITT effect from a randomized trial always requires no adjustment for confounding. **A1**: False. While baseline confounding is eliminated by randomization, loss to follow-up can create post-randomization selection bias requiring adjustment.

Q2: What makes a conventional "per-protocol analysis" (restricting to adherers) potentially biased? **A2**: Adherence is post-randomization and typically related to prognosis, creating confounding through $A \leftarrow U \rightarrow Y$. Simply restricting to adherers doesn't eliminate this bias.

Q3: Why might a per-protocol analysis be more informative than ITT for decision-making? **A3**: Decision-makers typically want to know the effect of actually following a treatment strategy, not the effect of being assigned to it with uncertain adherence patterns.

6 Common Pitfalls & Checks

⚠ Pitfall 1: Misunderstanding "Per-Protocol"

Wrong: "Per-protocol analysis just means restricting to people who adhered." **Right**: Per-protocol effect is a causal contrast requiring adjustment for confounding, just like any observational analysis.
Check: Did you adjust for factors predicting adherence and outcome? If not, your estimate is biased.

⚠ Pitfall 2: Immortal Time Bias

Wrong: Starting follow-up at treatment initiation for the treated group but at an earlier time for controls. **Right**: All groups must have the same time zero—when eligibility is met and strategies could be assigned.
Check: Can every individual die/experience the outcome at any time after time zero? If treatment group has "immortal time" before treatment, you have bias.

⚠ Pitfall 3: Censoring Medically-Indicated Treatment Changes

Wrong: Censoring anyone who stops treatment, including those who stop due to toxicity. **Right**: Realistic strategies include rules like "stop if toxicity occurs." Following this rule is adherence, not deviation.

Check: Does your protocol explicitly define when treatment changes are mandated? If stopping for toxicity is required, don't censor those patients.

Pitfall 4: Assuming ITT Is Always Conservative

Wrong: "The ITT effect is always closer to the null than the true effect, so it's safe." **Right**: In head-to-head trials or with high non-adherence, ITT can be further from the null than the per-protocol effect.
Check: Don't rely on ITT being conservative for safety assessment. Estimate per-protocol effects when evaluating harms.

Pitfall 5: Ignoring Time-Varying Confounding

Wrong: Using standard regression to adjust for baseline covariates when treatments and confounders vary over time. **Right**: Use g-methods (IP weighting, g-formula, g-estimation) when $A_{k-1} \rightarrow L_k \rightarrow A_k$.
Check: Draw a DAG. If past treatment affects future confounders, standard regression fails. Use g-methods.

! Essential Pre-Analysis Checks

Before analyzing any comparative study:

1. **Write out the target trial protocol** - What would the RCT look like?
2. **Define time zero precisely** - When is eligibility met?
3. **Specify treatment strategies exactly** - What actions at what times?
4. **Distinguish adherence from deviation** - What changes are mandated by the protocol?
5. **Identify time-varying confounders** - What post-baseline factors affect both treatment and outcome?
6. **Choose appropriate methods** - Do you need g-methods or will standard methods suffice?

7 Advanced Teaching Activities & Interactivity

7.1 Flipped Classroom: Protocol Critique

Pre-Class: Students read a published observational study and identify the implicit target trial.

In-Class: Small groups critique the study:

- What was the time zero?
- What strategies were compared?
- Were there time-zero errors?
- What confounders were adjusted for?
- Were g-methods needed?

Present findings to class and discuss.

7.2 Think-Pair-Share: When Is ITT Sufficient?

Prompt: "Your collaborator says: 'Randomization eliminates confounding, so we just need an ITT analysis. Why complicate things with per-protocol?' How do you respond?"

Think (2 min): Write down your response **Pair** (3 min): Discuss with a partner **Share** (10 min): Class discussion

Teaching Points:

- Randomization eliminates *baseline* confounding for effect of assignment
- Doesn't eliminate post-randomization selection bias (LTFU)
- Doesn't eliminate confounding for per-protocol effect
- ITT may not answer the policy question

7.3 Case Study: Hormone Therapy Controversy

Present the Women's Health Initiative randomized trial vs. observational studies of postmenopausal hormone therapy:

- Observational: Protective effect on CHD

- WHI Trial: Harmful effect on CHD

Group Activity: Analyze why estimates differed. Consider:

1. Different populations (healthy user bias)
2. Different time zeros (initiation at menopause vs. years later)
3. Different durations of follow-up
4. Different outcome ascertainment

Key Learning: Many "trial vs. observational study" discrepancies reflect poor emulation rather than fundamental limitations of observational data.

7.4 Interactive Simulation: Explore Non-Adherence Patterns

Provide R Shiny app where students manipulate:

- Degree of non-adherence (0-100%)
- Direction of confounding (high-risk more vs. less likely to adhere)
- Strength of treatment effect

Observe effects on:

- ITT estimate
- Naive per-protocol estimate
- Adjusted per-protocol estimate

Discussion: When does ITT fail to be conservative? When is adjustment most critical?

7.5 Retrieval Practice: Spaced Quizzing

Week 1: Define ITT and per-protocol effects **Week 3**: Identify time-zero errors in example studies **Week 5**: Determine when g-methods are needed **Week 7**: Design a target trial emulation

Each quiz includes:

- Multiple-choice questions
- Short-answer conceptual questions

- Code-reading exercises (identify errors in provided R code)

7.6 Peer Instruction: ConceptTests

Question 1: In a randomized trial with sustained treatment strategies, which statement is TRUE?

A. ITT analysis never requires adjustment for any variables B. Per-protocol analysis only requires adjustment for baseline confounders C. Both ITT and per-protocol analyses may require g-methods for time-varying factors D. Only observational studies need adjustment for time-varying confounding

Correct: C

Process:

1. Students vote individually (30 sec)
2. Discuss with neighbors (2 min)
3. Vote again
4. Instructor explains correct answer

Question 2: A study compares statin users to non-users. Statin users have lower CVD risk. Time zero was defined as the date of first statin prescription for users, and as a randomly selected date for non-users. What bias is present?

A. Immortal time bias B. Confounding by indication C. Selection bias D. All of the above

Correct: D

7.7 Problem-Based Learning: Design Your Own Target Trial

Scenario: You want to study whether intensive blood pressure control (target <120 mmHg) reduces dementia risk compared to standard control (<140 mmHg) in adults 60+.

Tasks:

1. Draft the full target trial protocol (6 components)

2. Identify potential eligibility times in observational data
3. Specify when you would censor clones
4. List time-varying confounders requiring g-methods
5. Anticipate challenges in emulation

Deliverable: 2-page protocol + 1-page analysis plan

Peer Review: Exchange with classmates for feedback

8 Appendix: Advanced Theory & Derivations

8.1 Formal Definition of Per-Protocol Effect Under Sustained Strategies

Let $\bar{A}_k = (A_0, A_1, \dots, A_k)$ denote treatment history through time k. A **treatment strategy** g is a function that assigns treatment A_k at time k based on past information:

$$g : (\bar{A}_{k-1}, \bar{L}_k) \mapsto \{0, 1\}$$

where $\bar{L}_k$ is covariate history through k. The strategy g can be:

- **Static**: $g(\bar{A}_{k-1}, \bar{L}_k) = a$ for all k (always treat or never treat)
- **Dynamic**: $g(\bar{A}_{k-1}, \bar{L}_k)$ depends on $\bar{L}_k$ (e.g., "treat if CD4 < 200")

Per-Protocol Effect: For strategies g and g', the per-protocol effect at time K is:

$$\Delta_{PP} = E[Y_K^g] - E[Y_K^{g'}]$$

where Y_K^g is the outcome at time K under strategy g.

Identification Conditions

To identify $E[Y_K^g]$ from observed data, we require:

1. **Consistency**: $Y_K = Y_K^{\bar{a}}$ when $\bar{A} = \bar{a}$
2. **Sequential exchangeability**: $Y_K^{\bar{a}} \perp!!! \perp A_k \mid \bar{A}_{k-1}, \bar{L}_k$ for all k
3. **Positivity**: $\Pr[A_k = g(\bar{A}_{k-1}, \bar{L}_k) \mid \bar{A}_{k-1}, \bar{L}_k] > 0$ for all k

Under these conditions:

$$E[Y_K^g] = \int E[Y_K \mid \bar{A} = \bar{a}^g, \bar{L}] \prod_{k=0}^{K-1} f(L_k \mid \bar{A}_{k-1}, \bar{L}_{k-1}) d\bar{L}$$

where $\bar{a}^g$ is the treatment path assigned by strategy g.

Proof Sketch: By sequential exchangeability and consistency, we can express the counterfactual mean as an iterated expectation over observed data distributions.

8.2 When Does ITT Equal Per-Protocol Effect?

Theorem: The ITT effect equals the per-protocol effect if and only if:

1. **Perfect adherence**: $A_k = g_z(k)$ for all k, where g_z is the strategy assigned by Z
2. **Exclusion restriction**: Z affects Y only through $\bar{A}$

Proof:

Under perfect adherence and exclusion restriction:

$$E[Y_K^{z=1}] = E[Y_K^{g_1}]$$

$$E[Y_K^{z=0}] = E[Y_K^{g_0}]$$

Therefore:

$$\Delta_{ITT} = E[Y_K^{z=1}] - E[Y_K^{z=0}] = E[Y_K^{g_1}] - E[Y_K^{g_0}] = \Delta_{PP}$$

Numeric Example:

Consider a 2-period trial ($k \in \{0, 1\}$, outcome at $k = 2$):

- Perfect adherence: All $Z = 1$ follow g_1: $(A_0 = 1, A_1 = 1)$; all $Z = 0$ follow g_0: $(A_0 = 0, A_1 = 0)$
- Exclusion restriction holds

Suppose true counterfactual risks:

$$\Pr[Y_2^{g_1} = 1] = 0.20$$

$$\Pr[Y_2^{g_0} = 1] = 0.30$$

Then observed risks:

$$\Pr[Y_2 = 1|Z = 1] = \Pr[Y_2^{g_1} = 1] = 0.20$$

$$\Pr[Y_2 = 1|Z = 0] = \Pr[Y_2^{g_0} = 1] = 0.30$$

Thus $\Delta_{ITT} = \Delta_{PP} = -0.10$.

8.3 Selection Bias from Loss to Follow-Up in ITT Analysis

Even with randomization, censoring C can induce selection bias for the ITT effect.

DAG:

$$Z \to A \to L \to C$$

$$L \to Y$$

Conditioning on $C = 0$ opens collider-stratification bias via $Z \to A \to L \leftarrow C$.

Formal Proof:

We want $E[Y^{z=1,c=0}] - E[Y^{z=0,c=0}]$, but observe:

$$E[Y \mid Z = 1, C = 0] - E[Y \mid Z = 0, C = 0]$$

These differ because:

$$E[Y \mid Z = z, C = 0] = \sum_{\ell} E[Y \mid Z = z, C = 0, L = \ell] \Pr[L = \ell \mid Z = z, C = 0]$$

The distribution $\Pr[L \mid Z = z, C = 0]$ differs from $\Pr[L^z]$ due to selection on C.

IP Weighting Solution:

Create weights:

$$W = \frac{1}{\Pr[C = 0 \mid Z, L]}$$

Then:

$$E\left[\frac{Y \cdot I(C = 0)}{\Pr[C = 0 \mid Z, L]} \mid Z = z\right] = E[Y^{z,c=0}]$$

Numeric Example:

Group	$\Pr[Y = 1 \mid C = 0]$	$\Pr[C = 0]$	True $E[Y^{z,c=0}]$
$Z = 1$	0.15	0.90	0.18
$Z = 0$	0.25	0.95	0.24

Naive estimate (complete cases): $0.15 - 0.25 = -0.10$ **True ITT effect**: $0.18 - 0.24 = -0.06$ **Bias**: -0.04 (underestimates benefit)

With IP weighting:

$$W_{Z=1} = \frac{1}{0.90}, \quad W_{Z=0} = \frac{1}{0.95}$$

Weighted estimate recovers -0.06.

8.4 Cloning + Censoring + Weighting: Formal Justification

Setup: At baseline, individual i's data are consistent with strategies $\{g_1, g_2, \dots, g_m\}$. We cannot uniquely assign them to one strategy.

Procedure:

1. Create m clones $\{i_1, i_2, \dots, i_m\}$, assign each to one strategy
2. At each time k, censor clone i_j if observed treatment A_k violates strategy g_j
3. Weight by inverse probability of remaining uncensored

Key Result: If censoring at time k depends on $(\bar{A}_{k-1}, \bar{L}_k)$, and we weight by:

$$W_k = \prod_{j=0}^{k-1} \frac{1}{\Pr[C_{j+1} = 0 \mid \bar{A}_j, \bar{L}_j, C_j = 0]}$$

Then:

$$E\left[\frac{Y_K \cdot I(\bar{C} = \bar{0})}{\prod_k W_k} \mid g\right] = E[Y_K^g]$$

Proof Sketch: By sequential exchangeability of censoring:

$$E[Y_K^{g,\bar{c}=\bar{0}}] = E\left[E[Y_K \mid \bar{A} = \bar{a}^g, \bar{C} = \bar{0}, \bar{L}] \prod_k \Pr[C_{k+1} = 0 \mid \bar{A}_k, \bar{L}_k]\right]$$

The weight $1/\Pr[C_{k+1} = 0 \mid \cdots]$ removes the selection.

Numeric Example:

Two strategies: g_1 (immediate treatment), g_2 (delay treatment).

Person 1 delays treatment:

- Clone 1 (assigned g_1): Censored at $k = 0$ when treatment not initiated

  - Weight at censoring: $W_0 = 1/\Pr[\text{initiate} \mid L_0] = 1/0.30 = 3.33$

- Clone 2 (assigned g_2): Remains uncensored
  - If dies at $k = 5$: Contributes death to g_2 strategy
  - Weight: $W_5 = \prod_{k=0}^{4} 1/\Pr[C_{k+1} = 0 \mid \bar{A}_k, \bar{L}_k]$

Person 2 initiates immediately, then stops at $k = 2$ due to toxicity (mandated by g_1):

- Clone 1 (assigned g_1): Remains uncensored (stopping is adherence)
- Clone 2 (assigned g_2): Censored at $k = 0$ when treatment initiated

8.5 Bias Decomposition for Naive Per-Protocol Analysis

Setting: Randomized trial with non-adherence. True per-protocol effect Δ_{PP}, observed naive estimate $\hat{\Delta}_{\text{naive}}$ comparing $A = 1$ vs $A = 0$ without adjustment.

Bias:

$$\text{Bias} = E[\hat{\Delta}_{\text{naive}}] - \Delta_{PP}$$

Can be decomposed:

$$\text{Bias} = \underbrace{\text{Confounding bias}}_{\text{Selection into } A} + \underbrace{\text{Selection bias}}_{\text{Censoring/LTFU}}$$

Confounding bias:

$$E[Y \mid A = 1] - E[Y \mid A = 0] = \underbrace{E[Y^{a=1}] - E[Y^{a=0}]}_{\text{Causal effect}} + \underbrace{E[Y^{a=0} \mid A = 1] - E[Y^{a=0} \mid A = 0]}_{\text{Confounding}}$$

Selection bias (if conditioning on $C = 0$):

$$E[Y \mid A = a, C = 0] \neq E[Y^{a,c=0}]$$

Numeric Example:

Suppose unmeasured U (high-risk):

U	$\Pr[U]$	$\Pr[A = 1 \mid U]$	$E[Y^{a=1} \mid U]$	$E[Y^{a=0} \mid U]$
1	0.5	0.7	0.20	0.60
0	0.5	0.3	0.10	0.30

True PP effect:

$$\Delta_{PP} = 0.5(-0.40) + 0.5(-0.20) = -0.30$$

Naive estimate:

$$\Pr[A = 1] = 0.5(0.7) + 0.5(0.3) = 0.50$$

$$E[Y \mid A = 1] = \frac{0.5(0.7)(0.20) + 0.5(0.3)(0.10)}{0.50} = 0.155$$

$$E[Y \mid A = 0] = \frac{0.5(0.3)(0.60) + 0.5(0.7)(0.30)}{0.50} = 0.285$$

$$\hat{\Delta}_{\text{naive}} = 0.155 - 0.285 = -0.130$$

Bias: $-0.130 - (-0.30) = +0.17$ (attenuation toward null)

Confounding component:

$$E[Y^{a=0} \mid A = 1] - E[Y^{a=0} \mid A = 0] = 0.515 - 0.315 = +0.20$$

This positive confounding (high-risk more likely treated) attenuates the beneficial effect.

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